

Achieving Long-Lived Visible-Light-Excited Phosphorescence Through Orchestrating Charge Transfer in Terpyridine-Based Donor–Acceptor Fluorophores

Jiayin Zhou, Junjuan Shi, Lijun Mao,* Yu Wang, Wuzhen Luo, Min Qi, Longqiang Li, Ming Wang,* Guangqiang Yin,* and Tao Chen*

Long-lived luminescent materials have garnered considerable attention due to their fascinating self-sustained afterglow emission and promising applications. However, most afterglow materials remain constrained to high-energy excitation, inducing accelerated photodegradation and posing biosafety risks through mutagenic effects. Herein, the study presents a molecular engineering paradigm in low-energy excitation to realize long-lived visible-light-excitable phosphorescence (VEP) via controllably modulating charge transfer (CT) processes in donor-acceptor (D-A) designed fluorophores. By anchoring electron-donating triphenylamine (TPA) to a π -accepting terpyridine (Tpy) unit, the D-A engineered fluorophores serve as versatile platforms for introducing dynamic Zn(II)-Tpy coordination and sequential protonation to largely enhance CT for bathochromic absorption shift, thereby enabling tunable and multicolor VEP. Furthermore, the photo-oxidation process induces structural evolutions of fluorophores to modulate intramolecular CT (ICT), resulting in ultralong VEP lasting more than 3 s (artificial sunlight excitation) and 1 s (low-power LED) under ambient conditions. Encouraged by the tunable properties of these versatile VEP materials, time-gated anti-counterfeiting and multi-level encryption are demonstrated with a high security level. This work establishes an effective toolbox for designing and fabricating VEP materials through precise CT engineering, advancing their adaptability for next-generation photonic technologies.

self-sustained emission.^[1,2] Benefiting from their unique features, such as long emission lifetimes and large Stokes shifts, long-lived luminescent materials have exhibited great potential in bioimaging,^[3–8] chemical sensing,^[9–12] optoelectronic devices,^[13–16] and so on. Since the creation of artificial afterglow materials, the excitation sources of both inorganic persistent phosphors and organic/organometallic phosphorescence materials have been predominantly limited to high-energy ultraviolet (UV) light or radiation (X-, α -, β - or γ -rays, etc.).^[17–23] With no doubt, the high-energy irradiation induces accelerated material photoaging, while in biomedical contexts, such exposure may provoke severe biomolecular damage and mutagenic effects. Developing rational design methodologies to orchestrate the fabrication of lower-energy irradiation-excitable afterglow materials is highly desirable for enriching luminescent properties and expanding practical applications.

In contrast to high-energy irradiation sources, visible light (380–780 nm) exhibits enhanced operational accessibility, markedly reduced phototoxic risk in biological systems, and superior tissue penetrability. In recent years, rapid development and multifunctional applications of visible-light-excited phosphorescence (VEP) have been witnessed in diverse fields.^[24] Several cornerstone strategies that

1. Introduction

Long-lived luminescent materials have garnered widespread attention and demonstrated extensive applications owing to their

biological systems, and superior tissue penetrability. In recent years, rapid development and multifunctional applications of visible-light-excited phosphorescence (VEP) have been witnessed in diverse fields.^[24] Several cornerstone strategies that

J. Zhou, Y. Wang, W. Luo, M. Qi, L. Li, G. Yin, T. Chen
State Key Laboratory of Advanced Marine Materials
Ningbo Institute of Materials Technology and Engineering
Chinese Academy of Sciences
Ningbo 315201, P. R. China
E-mail: yguangqiang@nimte.ac.cn; tao.chen@nimte.ac.cn

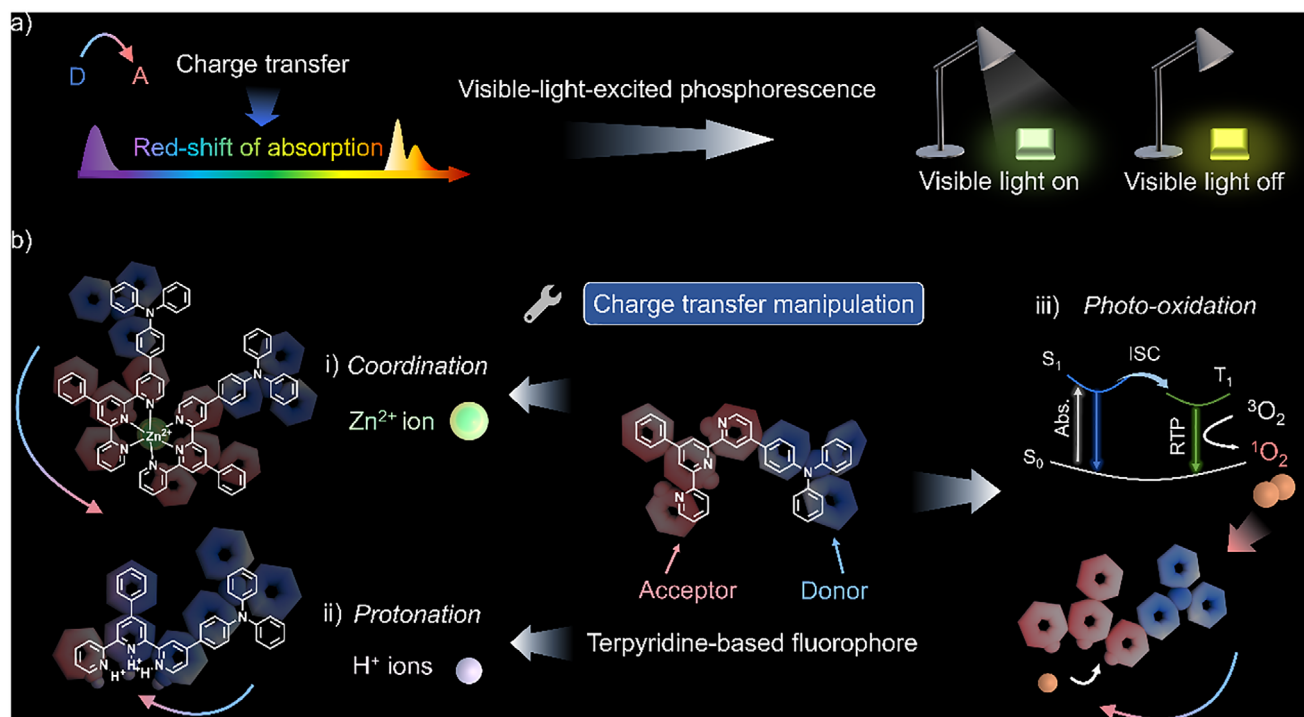
J. Zhou, Y. Wang, M. Qi, L. Li, G. Yin, T. Chen
School of Chemical Sciences
University of Chinese Academy of Sciences
Beijing 100049, P. R. China

J. Shi, M. Wang
State Key Laboratory of Supramolecular Structure and Materials
College of Chemistry
Jilin University
Changchun, Jilin 130012, P. R. China
E-mail: mingwang358@jlu.edu.cn

L. Mao
School of Pharmaceutical and Chemical Engineering & Institute for Advanced Studies
Taizhou University
Taizhou 318000, P. R. China
E-mail: ljmao@tzc.edu.cn

The ORCID identification number(s) for the author(s) of this article can be found under <https://doi.org/10.1002/adom.202501684>

DOI: 10.1002/adom.202501684



Scheme 1. Constructing long-lived visible-light-excited phosphorescence through manipulating charge transfer in terpyridine-based donor-acceptor fluorophores. a) The schematic illustration of realizing visible-light-excited phosphorescence by manipulating charge transfer to enable redshift of absorption. b) The illustration of modulating charge transfer of terpyridine-based D-A fluorophores, including (i) coordination, (ii) protonation, and (iii) photo-oxidation.

enabled a bathochromic absorption shift to the visible region have been established for achieving VEP materials, such as supramolecular assembly and aggregation,^[25–29] charge-transfer (CT) engineering,^[30,31] direct triplet state excitation,^[32–34] and carbon dots-based heterostructure design.^[35–38] Among these feasible pathways, the CT engineering based on abundant donor (D)-acceptor (A) molecular design has emerged as a more predictive paradigm for VEP by leveraging orbital hybridization manipulation and intersystem crossing (ISC) enhancement.^[39–43] For instance, Li et al. engineered a multibranch D-A architecture synergistically integrating intra- and intermolecular CT, achieving flashlight-excitable room-temperature phosphorescence (RTP) with a lifetime up to 344 ms.^[39] Also, An and coworkers developed VEP phosphors with a D-A design that leverages a σ -aromatic carboranyl cage to enhance intramolecular CT, enabling efficient excitation by 400–480 nm light and long-lived RTP with a lifetime more than 0.6 s.^[30] However, the VEP phosphors reported to date exhibit limited tunability for further optimization of their properties post-synthesis. That is, the absence of precise control strategies for manipulating CT significantly limits the programmable modulation of VEP, thereby hindering its adaptation to emerging application demands.

In this study, we propose a feasible toolbox to controllably modulate the CT in D-A engineered organic fluorophores to realize the redshift of absorption, thereby achieving multi-color VEP materials with tunable optoelectronic characteristics (**Scheme 1a**). 2,2':6',2''-Terpyridine (Tpy) as a strong π -acceptor was covalently grafted with electron-donating triphenylamine (TPA) motif, yielding D-A designed organic fluorophores as CT

manipulation platforms (**Scheme 1b**).^[44–47] By harnessing dynamic Zn(II)-Tpy coordination to enable metal-to-ligand charge transfer (MLCT) (**Scheme 1b(i)**),^[48–50] the absorption wavelength of organic fluorophores exhibited a considerable bathochromic shift up to 550 nm, resulting in multicolor VEP varying from green to red. In addition, the intramolecular CT (ICT) of D-A designed organic phosphors could be further modulated and enhanced by sequential protonation (**Scheme 1b(ii)**), yielding tunable and improved VEP. More importantly, the photo-oxidation process served as a controllable handle to effectively modulate ICT through UV-triggered structural evolution (**Scheme 1b(iii)**), enabling the resultant films to exhibit intense green afterglow lasting over 3 s (artificial sunlight excitation) and 1 s (low-power LED) under ambient conditions. By leveraging these advantages, time-gated anti-counterfeiting and multi-level encryption were demonstrated with a largely improved security level. This work outlines new pathways to precisely modulate CT for realizing VEP with tunable photophysical properties.

2. Results and Discussion

2.1. Preparation and Characterization of Purely Organic RTP Films

The D-A designed fluorophores (LX: LA–LE) were synthesized by efficient Suzuki coupling and Kröhnke reactions (**Scheme S1**, Supporting Information),^[46] with their chemical structures summarized in Figure S1 (Supporting Information). All the fluorophores were thoroughly characterized by ¹H, ¹³C nuclear

magnetic resonance (NMR) and matrix-assisted laser desorption/ionization time-of-flight (MALDI-TOF) mass spectrometry (Figures S2–S6, Supporting Information and Experimental Section). The solutions of LX were colorless under daylight and displayed intense blue or cyan fluorescence under 365 nm UV illumination (Figure S12, Supporting Information).

With these five excellent fluorophores, ultralong room-temperature phosphorescence (RTP) films were efficiently fabricated by incorporating them (0.1 wt.%) into polymethyl methacrylate (PMMA), which served as a rigid matrix for largely suppressing non-radiative transitions and providing an effective quencher barrier to obtain long-lived RTP materials (see details in the supporting information). All the composite films revealed excellent transparency (Figure S13, Supporting Information) and displayed three main absorption peaks (Figure S15, Supporting Information). Take LC@PMMA as an example, its absorption peaks appeared at 223, 286, and 352 nm, which could be assigned to high-energy $\pi-\pi^*$ transition, locally excited $\pi-\pi^*$ transition, and intra/inter-ligand charge transfer (ICT), respectively.^[51] Except for the dual RTP bands of LE@PMMA peaking at 440 and 565 nm, the other composite films displayed a single RTP emission band ≈ 540 nm (Figure S16, Supporting Information). The emission–excitation mapping of these films manifested that the fluorescence was independent of the excitation wavelength (Figure S17, Supporting Information). All RTP films demonstrated decent quantum efficiency, varying from 49% to 72% (Table S1, Supporting Information). Owing to largely boosted ISC efficiency by N heteroatoms, the LX@PMMA films showed ultralong RTP emission with lifetimes spanning from 144.73 to 319.27 ms (Figure S18 and Table S2, Supporting Information), displaying light green or yellow afterglow emission ranging from 1 to 3 s after the removal of 365 nm excitation (Figure S19, Supporting Information). These results demonstrate that the organic phosphors with a D–A design are good candidates for fabricating ultralong RTP materials.

2.2. Modulating CT by Dynamic Coordination for Obtaining VEP

To explore whether VEP can be achieved by enhancing CT, the dynamic Tpy-Zn coordination was introduced to enable MLCT for redshift of absorption (Figure 1a; Scheme S2, Supporting Information). By regulating the molar ratio of LX and Zn^{2+} , two complexes denoted as LX-Zn (1:1 metal-to-ligand ratio) and $(\text{LX})_2\text{-Zn}$ (1:2 metal-to-ligand ratio) could be obtained. The complexes were completely characterized by ^1H , ^{13}C NMR and high-resolution electrospray ionization time-of-flight (ESI-TOF) mass spectrometry (Figures S7–S11, Supporting Information and Experimental Section). Different from the transparent LX@PMMA film, both LX-Zn@PMMA and $(\text{LX})_2\text{-Zn@PMMA}$ films showed a yellow color under daylight (Figure 1b; Figures S13 and S14, Supporting Information), suggesting considerable photophysical changes. For instance, the LC-Zn@PMMA and $(\text{LC})_2\text{-Zn@PMMA}$ films revealed cyan and green-yellow fluorescence emissions under 365 nm irradiation. Compared with an intense light green afterglow of LC@PMMA, only a dim afterglow could be observed for LC-Zn@PMMA and $(\text{LC})_2\text{-Zn@PMMA}$ films after removing 365 nm UV excitation, indicating the change of optimal excitation wavelength.

Importantly, these zinc complex-involved films could be excited by visible light (450 nm) owing to the emergence of MLCT absorption in the visible region (Figure 1c; Figure S15, Supporting Information). For example, the LC-Zn@PMMA and $(\text{LC})_2\text{-Zn@PMMA}$ films exhibited MLCT absorption bands spanning 375–550 nm, with distinct maxima at 416 and 419 nm, respectively. Such a significant bathochromic absorption was further confirmed by the time-dependent density functional theory (TD-DFT) (Figures S20 and S21, Supporting Information). As such, $(\text{LX})_2\text{-Zn@PMMA}$ films were visible light excitable, revealing prompt emission peaking from 530 to 585 nm and VEP varying from 550 to 590 nm (Figure S22, Supporting Information). All coordination-modulated films revealed satisfactory quantum efficiency under 450 nm excitation, ranging from 63% for $(\text{LA})_2\text{-Zn@PMMA}$ to 40.4% for $(\text{LE})_2\text{-Zn@PMMA}$ (Table S1, Supporting Information).

Take LC-Zn@PMMA and $(\text{LC})_2\text{-Zn@PMMA}$ as an example; they displayed similar prompt emission at 530 nm ($\lambda_{\text{ex}} = 450$ nm), except for the increased intensity of $(\text{LC})_2\text{-Zn@PMMA}$ (Figure 1d). Accordingly, the VEP emission of both zinc complex-involved films appeared at 534 nm, in addition to a broader emission band for $(\text{LC})_2\text{-Zn@PMMA}$ that could be observed (Figure 1e). After the fitting analysis, the lifetimes of these coordination-modulated films are several milliseconds (Figure 1f; Figure S23 and Table S2, Supporting Information). At 77 K, the LC-Zn@PMMA and $(\text{LC})_2\text{-Zn@PMMA}$ displayed significantly elongated VEP lifetimes of up to 1320.10 and 1336.69 ms (Figure S24, Supporting Information), eliminating the thermally activated delayed fluorescence (TADF). Notably, these coordination-modulated films could be excited by table LED and showed intense ultralong multicolor afterglow at 77 K (Figure 1g; Figures S25 and S26, Supporting Information). With the extension of π -conjugation, the bathochromic afterglow shifts from yellow for $(\text{LA})_2\text{-Zn@PMMA}$ to red for $(\text{LD})_2\text{-Zn@PMMA}$, $(\text{LD})_2\text{-Zn@PMMA}$, $(\text{LE})_2\text{-Zn@PMMA}$, and $(\text{LE})_2\text{-Zn@PMMA}$. In short, these results demonstrate that VEP can be realized by enhancing the CT of D-A-engineered fluorophores.

2.3. Enhancing ICT by Sequential Protonation to Access VEP

Encouraged by leveraging coordination to boost CT for achieving VEP, we supposed that the related RTP performance could be further improved and controllably regulated by sequential protonation. As shown in Figure S27a (Supporting Information), the protonation of the D-A designed fluorophores showed an apparent color change from colorless to red, indicating a prominent bathochromic shift in absorption after protonation (Figure S27b, Supporting Information). As a result, the protonated molecule-related films exhibited a yellow color under daylight (Figure S27c, Supporting Information). To gain more insight into the protonation effect, LC was taken as a model fluorophore. By regulating the stoichiometric ratios, monoprotonated, diprotonated, and triprotonated LC can be efficiently generated,^[52] which are denoted as LC-1H, LC-2H, and LC-3H, respectively. The sequential protonation was confirmed by ^1H NMR spectra, showing obvious downfield shifts of protons owing to the loss of electron density (Figure 2). Especially, pronounced downfield shifts observed for pyridinyl protons unambiguously confirm the protonation of Tpy

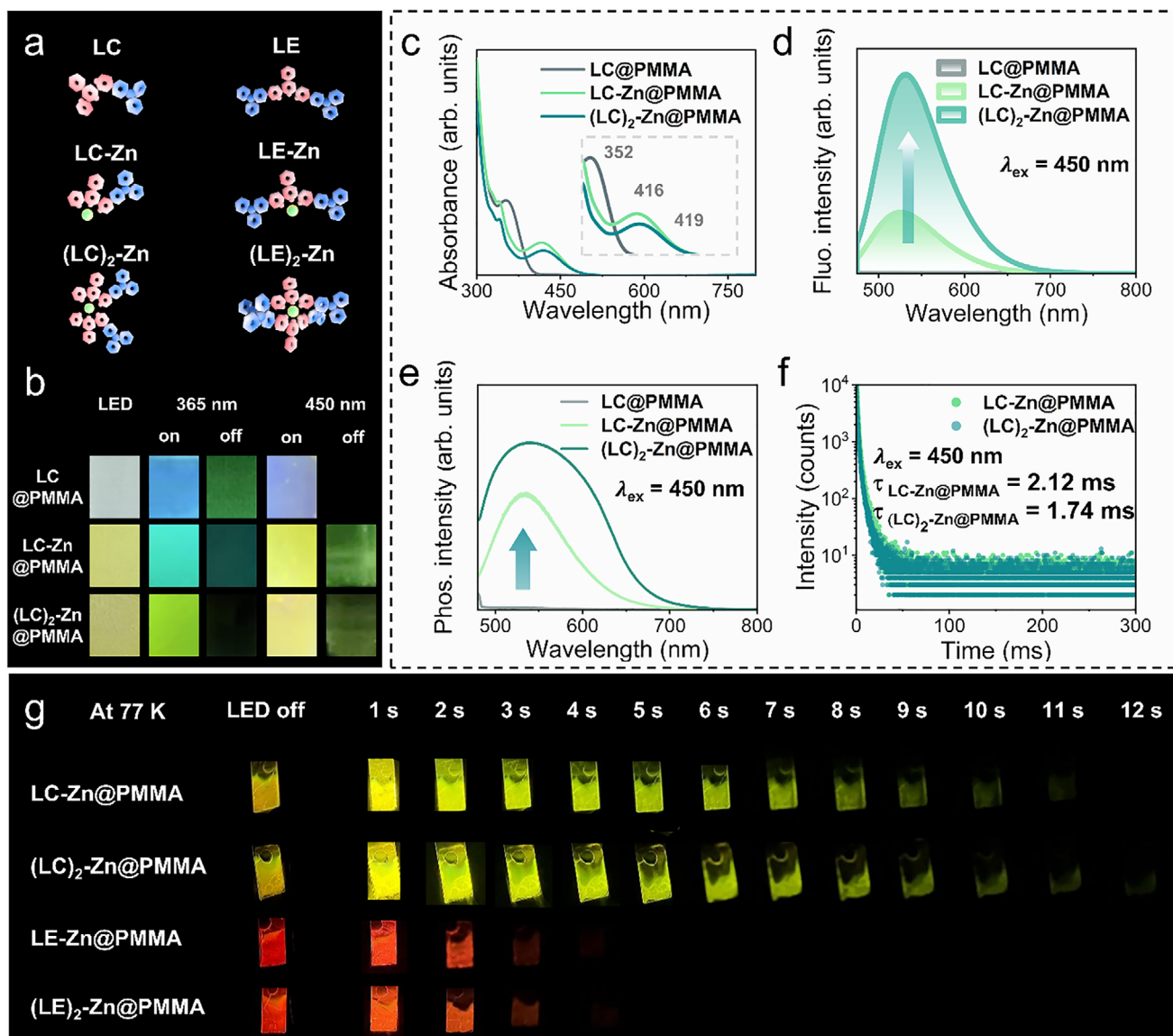


Figure 1. The photophysical characterization of pristine and coordinated films. a) The schematic representation of the LC, LC-Zn, (LC)₂-Zn, LE, LE-Zn, and (LE)₂-Zn. b) The photographs of the composite films captured under white table LED, during 365 nm UV exposure, and after cessation of 365 and 450 nm irradiation at room temperature. c) UV-vis absorbance spectra, d) fluorescence spectra, and e) RTP spectra of LC@PMMA, LC-Zn@PMMA, and (LC)₂-Zn@PMMA, $\lambda_{ex} = 450$ nm. f) Time-resolved emission-decay profiles of LC-Zn@PMMA and (LC)₂-Zn@PMMA. g) The afterglow photographs of the composite films captured after the cessation of the white table LED at 77 K.

units. For instance, the 3'-Tpy proton shows a significant downfield shift from 8.90 to 9.20 ppm after complete protonation of pyridine subunits due to the electron deshielding effect.

Thereafter, these sequentially protonated phosphors were incorporated into the PMMA matrix to fabricate visible-light-excited RTP films. In contrast to transparent LC@PMMA, the LC-1H@PMMA, LC-2H@PMMA, and LC-3H@PMMA displayed an increased yellow color under daylight (Figure S28a, Supporting Information). The corresponding films demonstrated ultralong RTP emission with lifetimes varying from 194 to 226 ms (Figure S28b, Supporting Information). Upon exposure to 450 nm light, the LC-1H@PMMA, LC-2H@PMMA, and

LC-3H@PMMA exhibited green, yellowish, and orange colors, respectively. After the removal of 450 nm light, all the composite films showed yellow VEP lasting over 0.2 s, indicating that the protonation had a minor impact on the energy level of the triplet excited state. In addition, the protonated LC-related films could be excited by table LED, displaying orange afterglow emission over 5 s (LC-1H@PMMA), 6 s (LC-2H@PMMA), and 7 s (LC-3H@PMMA) at 77 K (Figure 3b).

Excitation spectra were recorded to figure out the visible-light excitation of protonated LC-involved films (Figure 3c). In contrast to LC@PMMA, the excitation spectra of LC-1H@PMMA, LC-2H@PMMA, and LC-3H@PMMA displayed a broad band in the visible spectral region (400–500 nm, $\lambda_{max} = 430$ nm) ascribed to

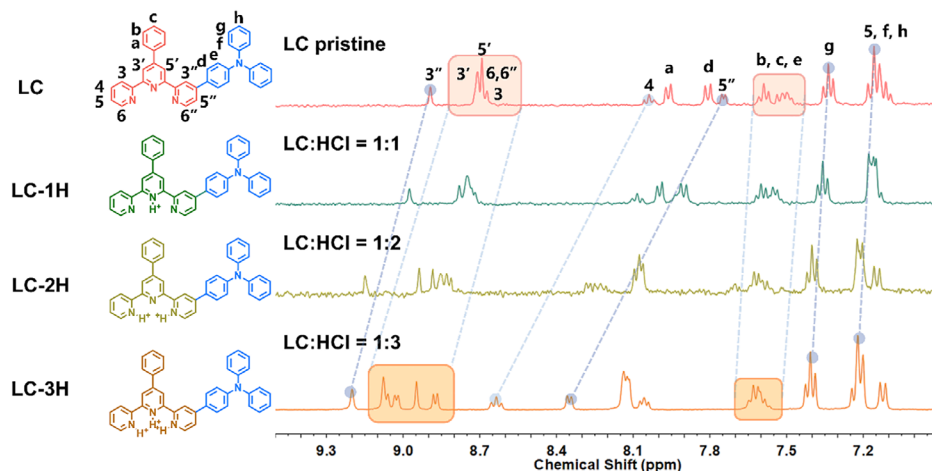


Figure 2. The ^1H NMR spectra (aromatic region) of LC before and after treating with onefold, twofold, and threefold molar HCl in methanol- d_4 .

the enhanced ICT, showing increased intensity with the increase of protonation degree. The UV–vis absorption of protonated LC-involved films followed the same increasing trend, demonstrating a sequential boosted ICT absorption in the visible spectral region (Figure S28a, Supporting Information). Such a considerable ICT absorption was further validated by theoretical calculations (Figures S29–S31, Supporting Information). Different from the pristine LC@PMMA film with only prompt emission at 420 nm, a new peak at 536 nm emerges for the protonated LC-involved films (Figure 3d). The corresponding films revealed an increased emission intensity at 536 nm with increasing protonation degree. Interestingly, the emission band of protonated LC-involved films was dependent on excitation wavelength, exhibiting a distinct red shift from blue to yellowish upon changing the excitation wavelength from 365 to 480 nm (Figure S32, Supporting Information). Particularly, the LC-3H@PMMA exhibited an obvious red shift in emission band from 410 to 550 nm, along with CIE coordinates changing from (0.22, 0.22) to (0.41, 0.57).

By using 450 nm excitation, one emission band peaking at 520 nm for LC-1H@PMMA and LC-2H@PMMA, 532 nm for LC-3H@PMMA could be observed (Figure 3d). In addition, these protonated LC-involved films displayed increased and slightly red-shifted VEP with increasing protonation degree, demonstrating 550 nm for LC-1H@PMMA and 558 nm for LC-3H@PMMA (Figure 3f). The VEP lifetimes of the corresponding films were comparable and calculated as 15.59 ms for LC-1H@PMMA, 12.58 ms for LC-2H@PMMA, and 11.57 ms for LC-3H@PMMA (Figure 3e; Table S2, Supporting Information). When cooling to 77 K, both prompt and VEP emission remained almost unchanged, apart from largely elongated lifetimes of 432.26 ms for LC-1H@PMMA, 613.35 ms for LC-2H@PMMA, and 867.83 ms for LC-3H@PMMA under 450 nm excitation (Figure S33, Supporting Information). The protonation of the LA, LB, LD, and LE also enables visible-light-excitable RTP with a maximum lifetime of 7.38 ms for LE-3H@PMMA by using 450 nm as an excitation source (Figures S34–S35, Supporting Information). Although protonation and coordination are generally reversible under alkaline conditions, the hydrophobic PMMA matrix enables these RTP films to maintain stable emission bands upon organic/inorganic base exposure (Figure S36, Supporting Infor-

mation). Taken together, the Tpy-functionalized design enables sequential protonation processes for controllably regulating CT, leading to multicolor VEP emission.

To gain more insights into the photophysical processes, density functional theory (DFT) and TD-DFT calculations were performed to determine the optoelectronic properties by using LC-related fluorophores as a model system (Figure 4). Considering the existence of Zn cations, geometries of the coordination complexes were optimized using the B3LYP/def2-SVP level of theory. As shown in Figure 4a, the highest occupied molecular orbitals (HOMOs) of LC, $(\text{LC})_2\text{-Zn}$, and LC-3H were mainly localized on TPA moieties, while the lowest unoccupied molecular orbitals (LUMOs) were mainly on the Tpy moieties, indicating the formation of CT characteristics for promoting the minimization of singlet-triplet splitting energy (ΔE_{ST}).^[53] Significantly, the separation of HOMO and LUMO was intensified by coordination and protonation. Moreover, the HOMO–LUMO energy gaps of $(\text{LC})_2\text{-Zn}$ (2.12 eV) and LC-3H (0.81 eV) were much lower than that of LC (3.96 eV), which implies a red-shift tendency of excitation wavelength.^[54,55] Besides, the natural transition orbitals (NTOs) analysis was performed to predict the effective transitions (Figure 4b). It was found that the transitions from $^1(n, \pi^*)$ to $^3(\pi, \pi^*)$ and $^1(\pi, \pi^*)$ to $^3(n, \pi^*)$ occurred for LC and LC-3H, demonstrating effective spin-orbit coupling (SOC) owing to good orbital overlap based on El-Sayed rule. Whereas, only transition from $^1(\pi, \pi^*)$ to $^3(\pi, \pi^*)$ occurred for $(\text{LC})_2\text{-Zn}$, indicating inefficient SOC due to poor orbital overlap. On the other hand, the ΔE_{ST} of $(\text{LC})_2\text{-Zn}$ and LC-3H were minimized to 0.05 and 0.17 eV, respectively, suggesting a more favorable ISC process compared with LC (0.75 eV). As such, the pristine fluorophores, Zn-incorporated complexes, and protonated fluorophores exhibit efficient ISC for populating triplet excitons.

2.4. Tuning CT via Aggregation and Photo-oxidation Induces Ultralong VEP

Generally, the aggregation of organic phosphors has a great impact on the optical properties.^[56–61] Therefore, we investigated the doping concentration-dependent photophysical properties. With

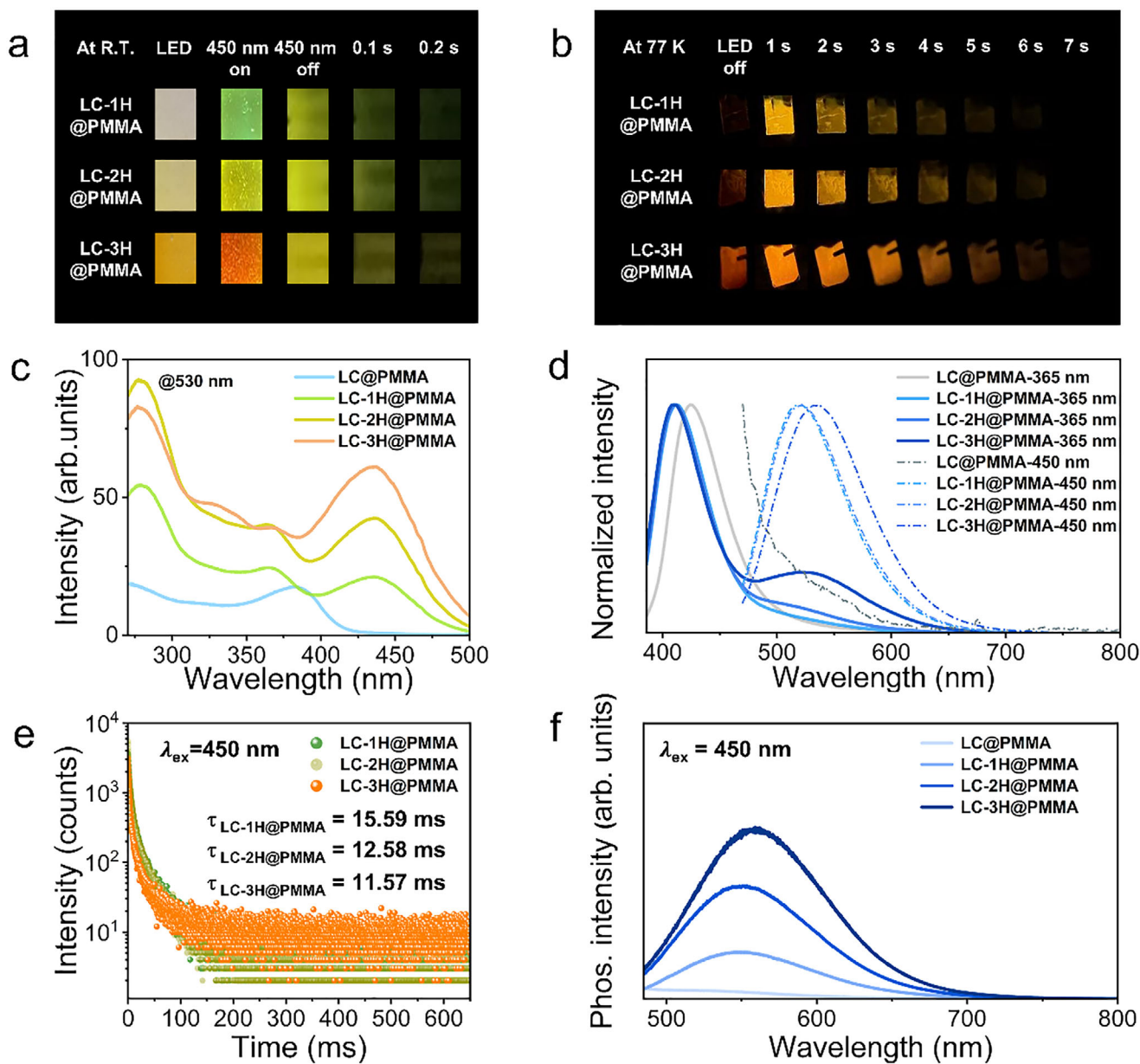


Figure 3. The photophysical characterization of protonated LC-related films. a) Photographs of LC-1H@PMMA, LC-2H@PMMA, and LC-3H@PMMA under table LED, 450 nm light, and at different time intervals after erasing 450 nm light irradiation at room temperature. b) Photographs of corresponding films show afterglow emission at different time intervals after erasing 450 nm light irradiation at 77 K. c) Excitation spectra of protonated LC-related films. d) Fluorescence spectra of composite films ($\lambda_{\text{ex}} = 365$ nm and 450 nm). e) Time-resolved emission-decay profiles of composite films ($\lambda_{\text{ex}} = 450$ nm). f) VEP spectra of composite films ($\lambda_{\text{ex}} = 450$ nm).

the increase of doping concentration from 0.1 to 5 wt.%, the absorption band originating from the locally excited π - π^* transition and ICT became structureless and enhanced along with the decrease of the absorption peaks at 223 nm owing to the aggregation of LC (Figure 5a). To figure out the molecular packing and interactions, X-ray-quality single crystals were cultured by slow diffusion of a dimethyl sulfoxide (DMSO) solution of LC (Figure 5j; Figure S37 and Table S3, Supporting Information). Taking Tpy unit as the basic plane, the dihedral angle between the substituted phenyl ring and Tpy is 42.32°, and the dihe-

dral angles between Tpy and the three planes of TPA are 23.77°, 47.25°, and 49.99°, respectively (Figure 5j(i)). Interestingly, J-aggregation with two modes of Tpy...Tpy and Tpy...TPA packing can be observed (Figure 5j(ii)). The stacking distance between adjacent molecules is ca. 3.51 Å, implying strong π - π interactions among Tpy units.^[62,63] The proximate intermolecular Tpy...TPA distance is $\approx$ 4.48 Å, which is conducive to through-space CT (TSCT) for promoting VEP.^[64] Although rod-like or needle-like crystals were obtained for LC-Zn, (LC)₂-Zn, and LC-3H (Figure S38, Supporting Information), they proved inadequate for

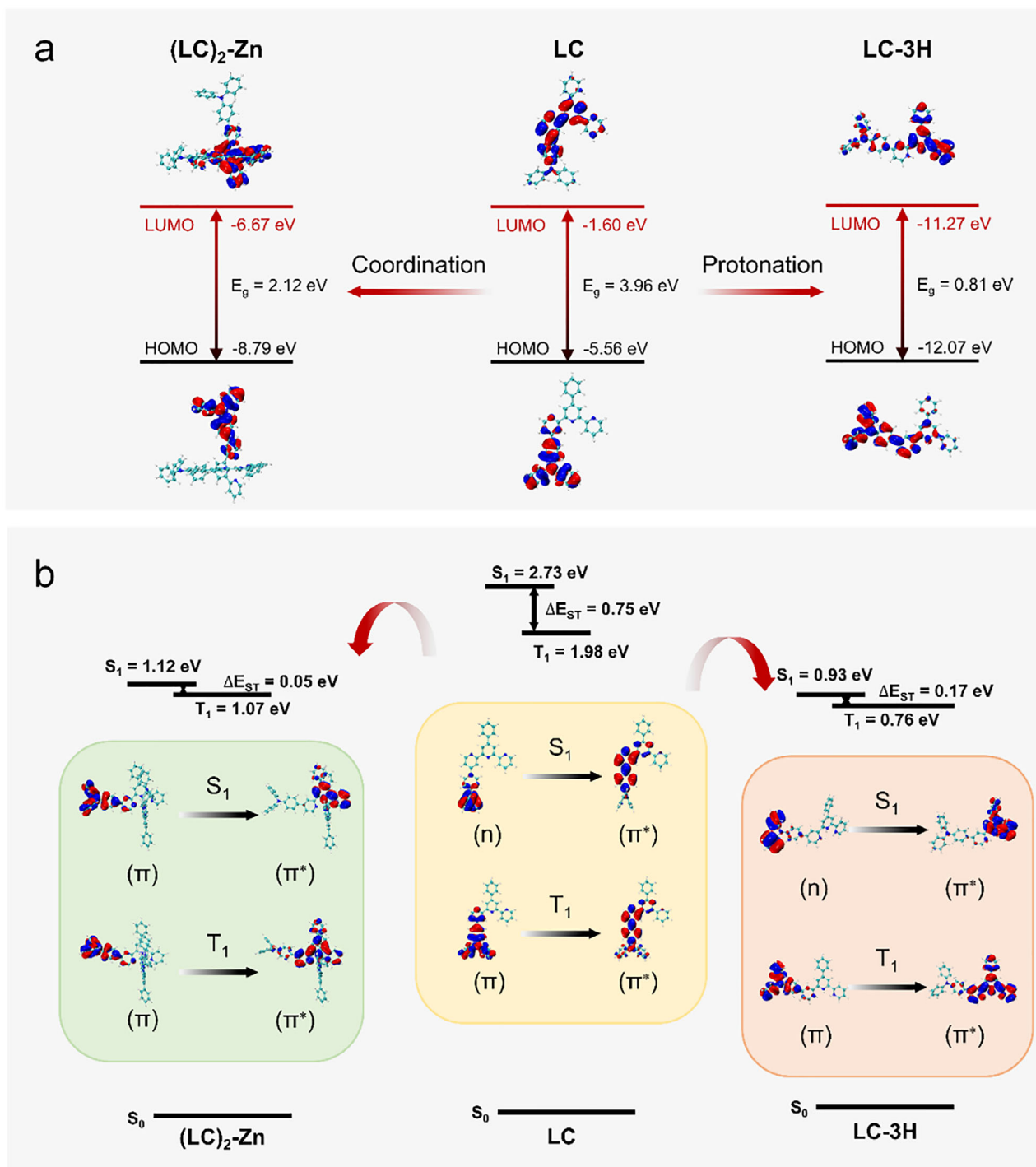


Figure 4. The DFT and TD-DFT calculations of LC, $(LC)_2-Zn$, and LC-3H. a) Calculated frontier molecular orbitals and orbital energies of LC, $(LC)_2-Zn$, and LC-3H. b) Vertical excitation energy levels and singlet-triplet splitting energies of LC, $(LC)_2-Zn$, and LC-3H with their natural transition orbitals (NTOs) for S_1 and T_1 and $n-\pi^*/\pi-\pi^*$ characteristic.

conducting X-ray diffraction analysis due to suboptimal crystal quality.

Notably, photochromic phenomena were observed for pristine films, which rendered ultralong VEP. After prolonged UV irradiation, the transparent LC@PMMA film changed into a

yellowish color (Figure S39, Supporting Information). As a result, the LC@PMMA film presents a prominent photochromic fluorescence changing from blue to light green, along with a green to yellow RTP transition after ceasing the excitation source (Figure S40, Supporting Information). To illustrate these

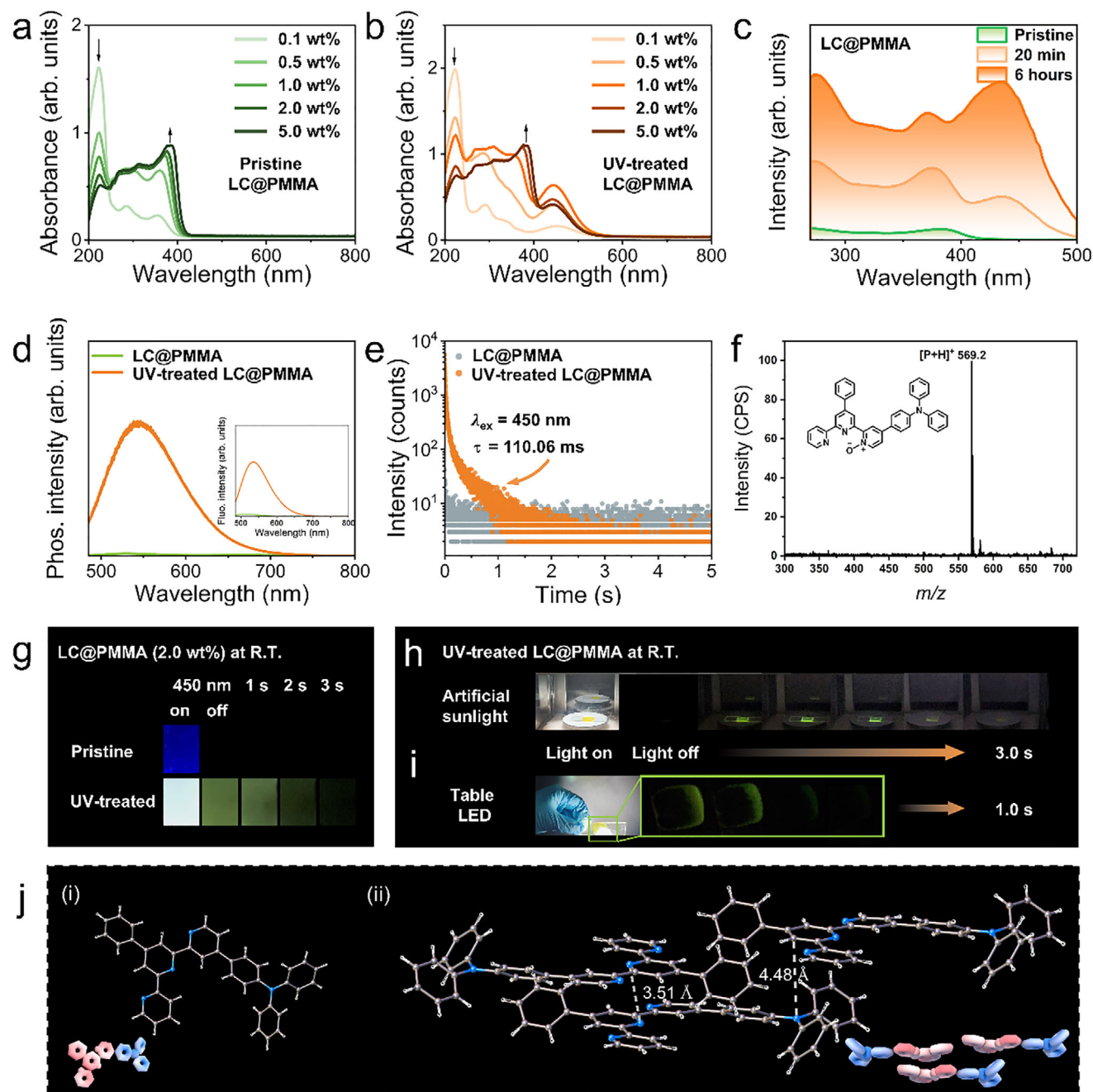


Figure 5. The UV–vis absorption spectra of LC@PMMA films (0.1–5.0 wt%), a) before and b) after UV irradiation for 1 hour (365 nm, 8 W). c) The excitation spectra of the LC@PMMA film (2.0 wt.%) under irradiation durations of 0, 20 min, and 6 h. d) RTP spectra, Inset: fluorescence spectra. e) Time-resolved emission-decay profile of pristine and UV-treated LC@PMMA films (2.0 wt.%). f) MS of UV-treated LC@PMMA film (2.0 wt.%). g) Photographs of pristine and UV-treated LC@PMMA films (2.0 wt.%) under 450 nm light and at different time intervals after erasing the irradiation at room temperature. h) Photographs of UV-treated LC@PMMA films (2.0 wt.%) under artificial sunlight excitation and at different time intervals after removing excitation at room temperature. i) Photographs of UV-treated LC@PMMA films (2.0 wt.%) under table LED (7 W) excitation and at different time intervals after removing excitation at room temperature. j) Single crystal structure of LC (i) and molecular packing diagram shows π – π stacking and TSCT distances (ii). Colors: N, blue; C, grey; H, white.

phenomena, we fully studied the corresponding photophysical properties (Figure 5). Importantly, the absorption band centred at 450 nm appeared after the UV treatment for 1 hour (Figure 5b), enabling the visible-light-excitable ability (Figure S41, Supporting Information). Moreover, the emission intensity underwent a

significant increase in the excitation spectra ranging from 270 to 500 nm after the continuous UV irradiation (Figure 5c), indicating the enhanced excitation efficiency.

Accordingly, the UV-treated LC@PMMA film could be efficiently excited by visible light (450 nm), displaying intense

fluorescence and VEP emissions at 536 and 543 nm, respectively (Figure 5d). Strikingly, the UV-treated LC@PMMA film revealed an ultralong lifetime of up to 110.06 ms upon being excited at 450 nm under ambient conditions. In sharp contrast to pristine LC@PMMA film, the UV-treated LC@PMMA film showed a green-yellow VEP lasting more than 3 s. More importantly, the UV-treated LC@PMMA film could be excited by artificial sunlight (Figure 5h) and low-power table LED (Figure 5i), displaying visible afterglow of up to 3 s and 1 s, respectively.

Such photochromic phenomena were irreversible in the long-term storage under ambient conditions and even when heated at 80 °C for several hours. No obvious change could be observed in the Fourier transform infrared (FTIR) (Figure S42a,b, Supporting Information), indicating that the molecular packing and interactions had no evident variation after UV irradiation. According to Li and Zhu's works,^[65,66] the formation of cation radicals may lead to distinct photochromism. Thus, electron paramagnetic resonance (EPR) analyses were carried out to probe the radicals. Interestingly, EPR results revealed that the radical signal of PMMA fragments ($g = 2.0023$)^[67–69] was significantly decreased after the introduction of LC (Figure S42c–f, Supporting Information), implying that the free radicals of PMMA were captured by LC molecules. As such, we suspected that new species might be generated in the photochromic process.

To probe the new species, we dissolved the UV-treated LC@PMMA film in DMSO- d_6 . It was found that no apparent change in the ^1H NMR for LC dissolved from the UV-treated film compared with pristine LC (Figure S43, Supporting Information), possibly due to the high detection limit of NMR spectroscopy. Thus, liquid chromatography-mass spectrometry (LC-MS) was further carried out to probe the trace species in the photochromic film. As expected, a series of signals, such as peaks at 569.2, 581.2, 583.2, 611.2, and 613.2, could be observed in the LC-MS (Figure 5f; Figure S44b, Supporting Information), which might be assigned to N-oxy-, dimethyl-, N-dioxy, methoxycarbonyl-, and N-trioxy-substituted LC species (Figure S44c, Supporting Information). The possible mechanism of such substitutions of LC is depicted in Figure S44a (Supporting Information), illustrating the formation of these new species owing to the generation of radical fragments of PMMA^[69] and oxidation by singlet oxygen. Generally, oxidation results in yellowing, which may be the cause of photochromism. Such photo-oxidation processes should lead to enhanced ICT to induce bathochromic absorption shift to the visible spectral band, ultimately contributing to ultralong VEP.

2.5. Applications of Visible-Light-Excitable RTP Materials

The tunable VEP materials are highly desirable for advanced anti-counterfeiting and information encryption applications.^[70–74] As shown in Figure 6a, we fabricated a time-gated anti-counterfeiting trademark by using LC@PMMA and LB@PMMA. The photochromism and its potential VEP not only enable high-level anti-counterfeiting but also provide a unique way to detect whether a fruit is exposed to sunlight during transportation. The trademark is originally transparent and gradually turns yellow after continuous light irradiation (Figure 6c). The “leaf” and “stalk” fabricated by LC@PMMA

changed into a yellowish color only after irradiation for 5 minutes, while the “orange” fabricated by LB@PMMA turned light yellow after 1 hour. After UV treatment, the trademark displayed blue “orange” with green “leaf” and “stalk” under 365 nm excitation, showing a vivid orange pattern lasting over 1.4 s after the removal of UV irradiation (Figure 6b). More importantly, the resultant trademark could be excited by visible light (450 nm) and exhibited a yellow afterglow with spatial-time evolution. As such, these photochromic and VEP materials hold great promise in anti-counterfeiting applications, demonstrating multiple collaborative paths to identify the authenticity of commodities.

Apart from anti-counterfeiting, we further demonstrated innovative multi-level encryption featuring “burn after reading” by integrating multiple VEP materials (Figure 6d). As a proof-of-concept, a dot-matrix encoding pattern (4×4) was designed and fabricated by using LC@PMMA, LB@PMMA, and $(\text{LD})_2\text{-Zn@PMMA}$. For simplicity, we defined the output of the emissive and the colored dots as “1”, while the non-emissive dots as “0”. Each row of dots displays four binary numbers, and the correct password is defined by transforming the corresponding binary numbers into a decimal number. Under ambient illumination, the cryptographic pattern resolves to the decimal numeral sequence “2004” – a temporal cipher referencing the establishment year of NIMTE (Ningbo Institute of Materials Technology and Engineering). Before UV-treatment, only the dots fabricated by $(\text{LD})_2\text{-Zn@PMMA}$ can be excited by 450 nm. Thus, only “0000” and “2004” can be decoded after removing 365 and 450 nm lights, respectively. Following 60 s of UV irradiation, the cryptographic matrix exhibited decrypted codes “0021” and “2025” upon deactivation of 365 and 450 nm excitation sources, respectively; these numerical ciphers correspond to the 21-year operational history and projected developmental milestones of NIMTE. Moreover, the encrypted information can be burned after reading by elongating the UV irradiation to turn on the VEP emission of LB@PMMA. Overall, these composite materials exhibit exceptional potential in time-gated anti-counterfeiting and multi-level information encryption, demonstrating a largely enhanced security level.

3. Conclusion

In summary, we demonstrated an effective toolbox to controllably manipulate the CT of D-A engineered organic fluorophores involving dynamic coordination, sequential protonation, and photo-oxidation, thereby realizing tunable VEP. Upon the introduction of dynamic Zn(II)-Tpy coordination and protonation, the molecular CT of organic phosphors was significantly enhanced, leading to a large bathochromic shift of absorption to the visible region. Consequently, efficient VEP materials were facilely achieved by incorporating the coordinated and protonated organic phosphors into the PMMA, demonstrating tunable and multicolor afterglow emission. Moreover, the photo-oxidation by UV irradiation induced structural evolution to effectively modulate ICT, endowing the resultant films with intense green VEP featuring ultralong lifetimes of more than 3 s (artificial sunlight excitation) and 1 s (low-power LED) under ambient conditions. By integrating these versatile VEP materials, time-gated anti-counterfeiting and multi-level encryption were

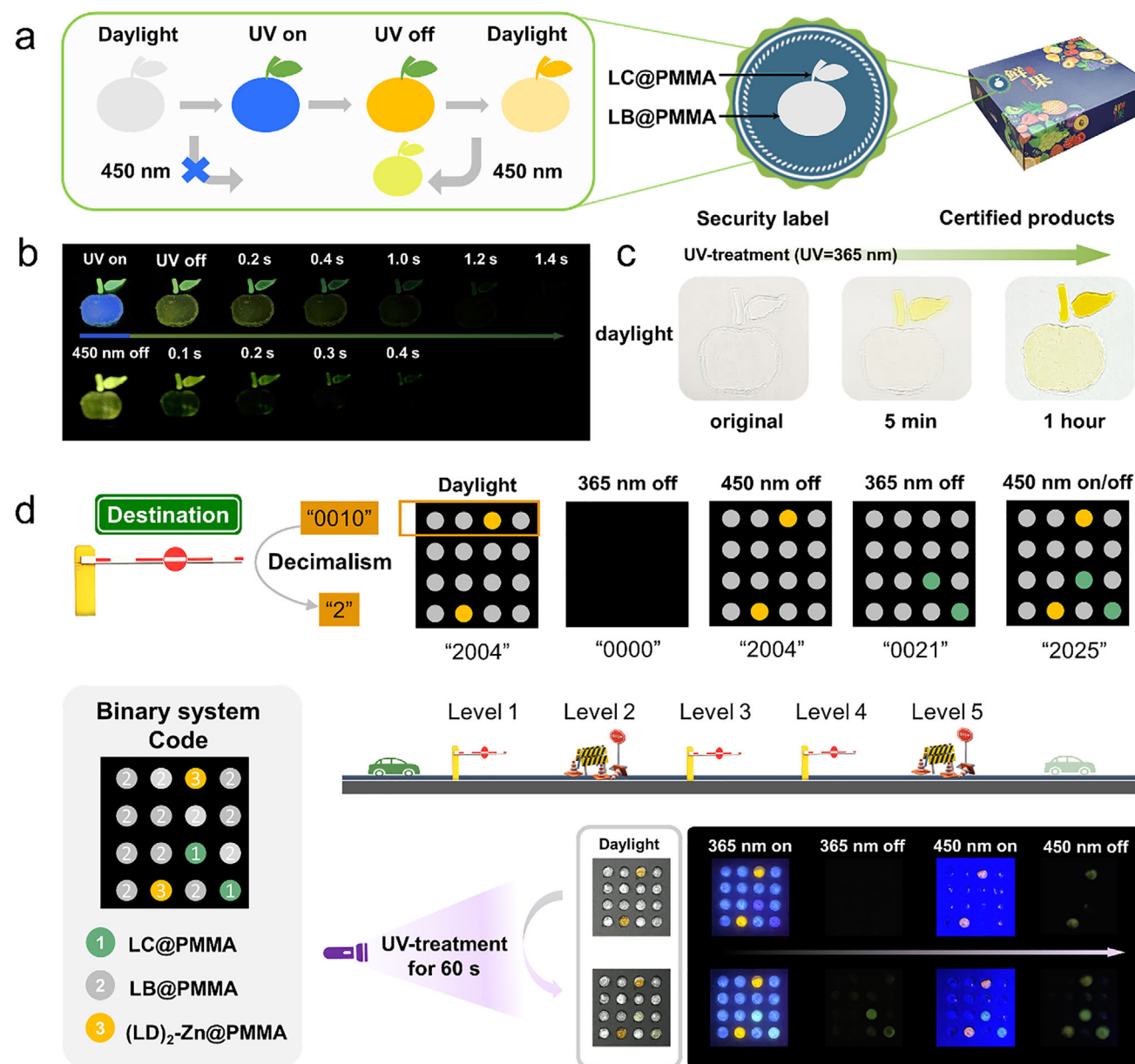


Figure 6. Time-gate anti-counterfeiting and multi-level encryption applications of VEP films. a) The schematic diagram illustrates the design of the time-gate anti-counterfeiting trademark. The "leaf" and "stalk" are fabricated by LC@PMMA, and the "orange" was fabricated by LB@PMMA. b) The fluorescence and phosphorescence photos of the anti-counterfeiting pattern. c) The photochromism of anti-counterfeiting trademarks upon continuous UV irradiation. d) The multi-level information encryption system was established by integrating LC@PMMA, LB@PMMA, and (LD)₂-Zn@PMMA.

demonstrated with a high security level. This study not only establishes novel strategies for achieving VEP through controllable manipulation of CT processes, but also advances the fundamental comprehension of photophysical processes governing optical materials, thereby paving the way for developing next-generation optoelectronic devices.

Supporting Information

Supporting Information is available from the Wiley Online Library or from the author.

Acknowledgements

J.Z. and J.S. contributed equally to this work. This project was financially supported by the Strategic Priority Research Program of the Chinese Academy of Sciences (No. XDB1210401), the National Natural Science Foundation of China (22205249, and 22401107), Zhejiang Provincial Natural Science Foundation of China (LQ23B040002), and the Ningbo Yongjiang Talent Programme (2024Z212).

Conflict of Interest

The authors declare no conflict of interest.

Data Availability Statement

Research data are not shared.

Keywords

amorphous phosphorescent film, charge transfer, data encryption, terpyridine, visible-light excitation

Received: May 25, 2025

Revised: July 26, 2025

Published online: September 5, 2025

- [1] W. Zhao, Z. He, B. Z. Tang, *Nat. Rev. Mater.* **2020**, 5, 869.
- [2] S. Xu, R. Chen, C. Zheng, W. Huang, *Adv. Mater.* **2016**, 28, 9920.
- [3] B. Chang, J. Chen, J. Bao, T. Sun, Z. Cheng, *Chem. Rev.* **2023**, 123, 13966.
- [4] J. Yang, Y. Zhang, X. Wu, W. Dai, D. Chen, J. Shi, B. Tong, Q. Peng, H. Xie, Z. Cai, Y. Dong, X. Zhang, *Nat. Commun.* **2021**, 12, 4883.
- [5] W. L. Zhou, Y. Chen, Q. Yu, H. Zhang, Z. X. Liu, X. Y. Dai, J. J. Li, Y. Liu, *Nat. Commun.* **2020**, 11, 4655.
- [6] J. Yang, X. Zhen, B. Wang, X. Gao, Z. Ren, J. Wang, Y. Xie, J. Li, Q. Peng, K. Pu, Z. Li, *Nat. Commun.* **2018**, 9, 840.
- [7] F. Ni, Z. Zhu, X. Tong, M. Xie, Q. Zhao, C. Zhong, Y. Zou, C. Yang, *Chem. Sci.* **2018**, 9, 6150.
- [8] D. Liu, W.-J. Wang, P. Alam, Z. Yang, K. Wu, L. Zhu, Y. Xiong, S. Chang, Y. Liu, B. Wu, Q. Wu, Z. Qiu, Z. Zhao, B. Z. Tang, *Nat. Photonics* **2024**, 18, 1276.
- [9] H. Hou, H. Wang, M. He, Q. Li, X. Wang, F. Guo, Q. Chen, L. Qu, C. Yang, *Angew. Chem., Int. Ed.* **2024**, 63, 202411323.
- [10] E. Hamzehpoor, C. Ruchlin, Y. Tao, C. H. Liu, H. M. Titi, D. F. Perepichka, *Nat. Chem.* **2023**, 15, 83.
- [11] Y. Zhou, W. Qin, C. Du, H. Gao, F. Zhu, G. Liang, *Angew. Chem., Int. Ed.* **2019**, 58, 12102.
- [12] S. G. Dunning, A. J. Nuñez, M. D. Moore, A. Steiner, V. M. Lynch, J. L. Sessler, B. J. Holliday, S. M. Humphrey, *Chem* **2017**, 2, 579.
- [13] R. Liu, C. Liu, C. Fu, Z. Zhu, K. Chen, C. Li, L. Wang, Y. Huang, Z. Lu, *Angew. Chem., Int. Ed.* **2024**, 63, 202312534.
- [14] T. Hua, X. Cao, J. Miao, X. Yin, Z. Chen, Z. Huang, C. Yang, *Nat. Photonics* **2024**, 18, 1161.
- [15] C. Lin, Z. Wu, H. Ma, J. Liu, S. You, A. Lv, W. Ye, J. Xu, H. Shi, B. Zha, W. Huang, Z. An, Y. Zhuang, R.-J. Xie, *Nat. Photonics* **2024**, 18, 350.
- [16] K. Chen, Y. Luo, M. Sun, C. Liu, M. Jia, C. Fu, X. Shen, C. Li, X. Zheng, X. Pu, Y. Huang, Z. Lu, *Angew. Chem., Int. Ed.* **2024**, 63, 202314447.
- [17] M. Dong, Z. Wang, Z. Lin, Y. Zhang, Z. Chen, Y. Wu, H. Ma, Z. An, L. Gu, W. Huang, *J. Am. Chem. Soc.* **2025**, 147, 4069.
- [18] X. Y. Dai, M. Huo, Y. Liu, *Nat. Rev. Chem.* **2023**, 7, 854.
- [19] Q. Liao, Q. Li, Z. Li, *Adv. Mater.* **2023**, 37, 2306617.
- [20] R. Gao, M. S. Kodaimati, D. Yan, *Chem. Soc. Rev.* **2021**, 50, 5564.
- [21] Z. Zhao, H. Zhang, J. W. Y. Lam, B. Z. Tang, *Angew. Chem., Int. Ed.* **2020**, 59, 9888.
- [22] J. Xu, S. Tanabe, *J. Lumin.* **2019**, 205, 581.
- [23] Y. Li, M. Gecevicius, J. Qiu, *Chem. Soc. Rev.* **2016**, 45, 2090.
- [24] L. Li, D. Liu, J. Zhou, M. Qi, G. Yin, T. Chen, *Mater. Horiz.* **2024**, 11, 5895.
- [25] D. Guo, W. Wang, K. Zhang, J. Chen, Y. Wang, T. Wang, W. Hou, Z. Zhang, H. Huang, Z. Chi, Z. Yang, *Nat. Commun.* **2024**, 15, 3598.
- [26] J. You, X. Zhang, Q. Nan, K. Jin, J. Zhang, Y. Wang, C. Yin, Z. Yang, J. Zhang, *Nat. Commun.* **2023**, 14, 4163.
- [27] X. Nie, J. Du, W. Huang, T. Wang, X. Wang, B. Chen, X. Zhang, G. Zhang, *Adv. Opt. Mater.* **2022**, 10, 2200099.
- [28] J. Wang, Z. Huang, X. Ma, H. Tian, *Angew. Chem., Int. Ed.* **2020**, 59, 9928.
- [29] S. Cai, H. Shi, J. Li, L. Gu, Y. Ni, Z. Cheng, S. Wang, W. W. Xiong, L. Li, Z. An, W. Huang, *Adv. Mater.* **2017**, 29, 1701244.
- [30] D. Tu, S. Cai, C. Fernandez, H. Ma, X. Wang, H. Wang, C. Ma, H. Yan, C. Lu, Z. An, *Angew. Chem., Int. Ed.* **2019**, 58, 9129.
- [31] X. F. Wang, H. Xiao, P. Z. Chen, Q. Z. Yang, B. Chen, C. H. Tung, Y. Z. Chen, L. Z. Wu, *J. Am. Chem. Soc.* **2019**, 141, 5045.
- [32] J. Yuan, Y. Wang, B. Zhou, W. Xie, B. Zheng, J. Zhang, P. Li, T. Yu, Y. Qi, Y. Tao, R. Chen, *Molecules* **2024**, 29, 1014.
- [33] J. Yuan, R. Chen, X. Tang, Y. Tao, S. Xu, L. Jin, C. Chen, X. Zhou, C. Zheng, W. Huang, *Chem. Sci.* **2019**, 10, 5031.
- [34] Y. Wang, Z. Zhang, L. Liu, S. Yuan, J. Ma, D. Liu, S. Xue, Q. Sun, W. Yang, *J. Mater. Chem. C* **2019**, 7, 9671.
- [35] B. Xu, Y. Jia, H. Ning, Q. Teng, C. Li, X. Fang, J. Li, H. Zhou, X. Meng, Z. Gao, X. Wang, Z. Wang, F. Yuan, *Small* **2024**, 20, 2304958.
- [36] H. Wu, Y. Kang, S. Jiang, K. Wang, L. Qu, C. Yang, *Small* **2024**, 20, 2402796.
- [37] Y. Huang, P. Li, *Chem. Eng. J.* **2024**, 480, 148157.
- [38] H. Shi, Z. Niu, H. Wang, W. Ye, K. Xi, X. Huang, H. Wang, Y. Liu, H. Lin, H. Shi, Z. An, *Chem. Sci.* **2022**, 13, 4406.
- [39] Y. Fan, S. Liu, M. Wu, L. Xiao, Y. Fan, M. Han, K. Chang, Y. Zhang, X. Zhen, Q. Li, Z. Li, *Adv. Mater.* **2022**, 34, 2201280.
- [40] S. Sun, L. Ma, J. Wang, X. Ma, H. Tian, *Natl. Sci. Rev.* **2022**, 9, nwab085.
- [41] Y. Wang, Q. Sun, L. Yue, J. Ma, S. Yuan, D. Liu, H. Zhang, S. Xue, W. Yang, *Adv. Opt. Mater.* **2021**, 9, 2101075.
- [42] Q. Xu, L. Ma, S. Sun, X. Ma, *J. Mater. Chem. C* **2021**, 9, 14623.
- [43] H. Bhatia, D. Ray, *J. Phys. Chem. C* **2019**, 123, 22104.
- [44] L. Duan, Q. Zheng, T. Tu, *Adv. Mater.* **2022**, 34, 2202540.
- [45] J. Zhou, D. Liu, L. Li, M. Qi, G. Yin, T. Chen, *Chin. Chem. Lett.* **2024**, 35, 109929.
- [46] S. Guan, Z. Wei, N. Han, Y. Wei, B. Xu, M. Wang, J. Shi, *Chin. Chem. Lett.* **2024**, 35, 109348.
- [47] H. Gu, X. Sun, Q. Zhao, H. Wang, X. Cheng, C. Yang, D. Qiu, *Molecules* **2023**, 28, 8027.
- [48] L. Duan, Q. Zheng, Y. Liang, T. Tu, *Adv. Mater.* **2024**, 36, 2409620.
- [49] G. Q. Yin, S. Kandapal, C. H. Liu, H. Wang, J. Huang, S. T. Jiang, T. Ji, Y. Yan, S. Khalife, R. Zhou, L. Ye, B. Xu, H. B. Yang, M. P. Nieh, X. Li, *Angew. Chem., Int. Ed.* **2021**, 60, 1281.
- [50] X. Yu, Y. Hu, C. Guo, Z. Chen, H. Wang, X. Li, *Supramol. Mater.* **2022**, 1, 100017.
- [51] X. Bi, Y. Pang, *J. Phys. Chem. B* **2016**, 120, 3311.
- [52] N. Yoshikawa, S. Yamabe, N. Kanehisa, T. Inoue, H. Takashima, *J. Phys. Org. Chem.* **2016**, 29, 269.
- [53] L. Sun, W. Hua, Y. Liu, G. Tian, M. Chen, M. Chen, F. Yang, S. Wang, X. Zhang, Y. Luo, W. Hu, *Angew. Chem., Int. Ed.* **2019**, 58, 11311.
- [54] G. Niu, X. Bi, Y. Kang, H. Zhao, R. Li, M. Ding, B. Zhou, Y. Zhai, X. Ji, Y. Chen, *Adv. Mater.* **2024**, 36, 2407199.
- [55] Q. Wang, C. Li, Y. Song, Q. Shi, H. Li, H. Zhong, J. Wang, F. Hu, *Chem. Sci.* **2023**, 14, 684.
- [56] G. Jiang, J. Yu, J. Wang, B. Z. Tang, *Aggregate* **2022**, 3, 285.
- [57] Y. Zhang, Y. Su, H. Wu, Z. Wang, C. Wang, Y. Zheng, X. Zheng, L. Gao, Q. Zhou, Y. Yang, X. Chen, C. Yang, Y. Zhao, *J. Am. Chem. Soc.* **2021**, 143, 13675.
- [58] Q. Li, Z. Li, *Acc. Chem. Res.* **2020**, 53, 962.
- [59] J. Yang, M. Fang, Z. Li, *Aggregate* **2020**, 1, 6.
- [60] M. Lian, Y. Mu, Z. Ye, Z. Lu, J. Xiao, J. Zhang, S. Ji, H. Zhang, Y. Huo, B. Z. Tang, *Aggregate* **2024**, 5, 560.
- [61] G. Yin, W. Lu, J. Huang, R. Li, D. Liu, L. Li, R. Zhou, G. Huo, T. Chen, *Aggregate* **2023**, 4, 344.
- [62] L. Sun, W. Zhu, W. Wang, F. Yang, C. Zhang, S. Wang, X. Zhang, R. Li, H. Dong, W. Hu, *Angew. Chem., Int. Ed.* **2017**, 56, 7831.

- [63] Y. Xie, Y. Ge, Q. Peng, C. Li, Q. Li, Z. Li, *Adv. Mater.* **2017**, 29, 1606829.
- [64] Y. Luo, Z. Pang, C. Li, K. Chen, X. Zheng, Y. Huang, Z. Lu, *J. Mater. Chem. C* **2020**, 8, 11603.
- [65] X. Xu, I. Sahalianov, H. Sun, Z. Li, S. Wu, B. Jiang, H. Agren, G. V. Baryshnikov, M. Zhang, L. Zhu, *Angew. Chem., Int. Ed.* **2025**, 64, 202422856.
- [66] Y. Yang, J. Wang, D. Li, J. Yang, M. Fang, Z. Li, *Adv. Mater.* **2021**, 33, 2104002.
- [67] H. Y. Kaptan, L. Tatar, *J. Appl. Polym. Sci.* **1997**, 65, 1161.
- [68] C. S. Bilen, D. J. Morantz, *Polymer* **1976**, 17, 1091.
- [69] G. Geuskens, C. David, *Die Makromol. Chem.* **1973**, 165, 273.
- [70] Q. Chen, L. Qu, H. Hou, J. Huang, C. Li, Y. Zhu, Y. Wang, X. Chen, Q. Zhou, Y. Yang, C. Yang, *Nat. Commun.* **2024**, 15, 2947.
- [71] S. Zhang, G. Liu, J. Chen, Y. Zhang, Q. Sun, S. Xue, W. Yang, *Adv. Opt. Mater.* **2023**, 11, 2301147.
- [72] J. Yu, Z. Sun, H. Ma, C. Wang, W. Huang, Z. He, W. Wu, H. Hu, W. Zhao, W. H. Zhu, *Angew. Chem., Int. Ed.* **2023**, 62, 202316647.
- [73] X. Wang, H. Ma, M. Gu, C. Lin, N. Gan, Z. Xie, H. Wang, L. Bian, L. Fu, S. Cai, Z. Chi, W. Yao, Z. An, H. Shi, W. Huang, *Chem. Mater.* **2019**, 31, 5584.
- [74] M. Louis, H. Thomas, M. Gmelch, A. Haft, F. Fries, S. Reineke, *Adv. Mater.* **2019**, 31, 1807887.